Feature Analysis and Business Intelligence Report

1. Introduction

The objective of this analysis was to evaluate the quality and business relevance of the engineered customer features generated from the Olist E-Commerce dataset. The resulting customer feature warehouse contains aggregated customer-level metrics including purchasing behavior, delivery performance, review patterns, seller diversity, product diversity, and payment behavior.

This analysis was conducted before applying machine learning models to understand feature distributions, identify data quality issues, detect outliers, and uncover business insights that could guide subsequent modeling decisions.

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2. Missing Value Analysis

A review of the engineered feature set revealed that most features were complete and suitable for machine learning applications.

Missing Features

Feature	Missing Values
avg_delivery_time_days	~2087
avg_delay_days	~2087

The missing values are likely associated with orders that were not successfully delivered or where delivery information was unavailable.

Implications

- Missing values are limited to delivery-related metrics.
- The proportion of missing data is relatively small compared to the overall customer population.
- Median imputation can be used during preprocessing.
- Missing delivery information may itself contain useful business information and can later be represented through indicator variables.

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3. Distribution Analysis

Recency Days

The recency distribution shows a moderate positive skew.

Key observations:

- Most customers purchased within the last 50–500 days.
- A small group of customers has been inactive for a very long period.
- The dataset contains both active and dormant customers.

Business Interpretation:

Customers with high recency values may represent potential churn candidates.

Frequency

The frequency distribution is highly concentrated around one purchase.

Key observations:

- Most customers placed only a single order.
- Repeat purchasing behavior is relatively uncommon.

Business Interpretation:

Customer retention appears to be a major challenge and should be explored further through segmentation.

Monetary Value

The monetary value distribution exhibits extreme right skewness.

Key observations:

- Most customers spend relatively small amounts.
- A small number of customers generate very large revenues.
- Several customers spend thousands of currency units above the average.

Business Interpretation:

The customer base follows a Pareto-like pattern where a small percentage of customers contributes a significant share of revenue.

Average Review Score

The distribution is concentrated around high ratings.

Key observations:

- Most customers provide ratings of 4 or 5.
- Negative reviews are comparatively rare.

Business Interpretation:

Overall customer satisfaction is high, although negative review behavior remains important for identifying at-risk customer groups.

Delivery Time

Delivery times are positively skewed.

Key observations:

- Most orders are delivered within a reasonable time frame.
- Some deliveries take significantly longer than average.

Business Interpretation:

Delivery performance may influence customer satisfaction and future purchasing behavior.

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4. Outlier Analysis

Outlier detection was performed on monetary value using the IQR method.

Results

- Approximately 7,841 customer records were identified as spending outliers.
- Several customers spend substantially more than the typical customer.

Business Interpretation

These customers likely represent:

- Premium customers
- High-value customers
- Business buyers
- Loyal repeat purchasers

Recommendation

These outliers should not be removed because they represent genuine business value rather than erroneous data.

For machine learning purposes:

- Apply log transformation to monetary value.
- Use robust scaling techniques when appropriate.

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5. Correlation Analysis

The correlation matrix was examined to understand relationships between engineered features.

Strongest Positive Correlations with Monetary Value

Feature	Correlation
avg_installments	0.31
unique_products	0.06
avg_delivery_time_days	0.06
bad_review_rate	0.05
unique_sellers	0.04

Key Findings

Installment Usage

Customers who spend more tend to use installment payments more frequently.

Business Implication:

Higher-value purchases are commonly financed through installments.

Product Diversity

Customers purchasing a wider variety of products generally spend more.

Business Implication:

Cross-selling opportunities may increase customer value.

Seller Diversity

Customers interacting with multiple sellers exhibit slightly higher spending behavior.

Business Implication:

Marketplace engagement contributes positively to revenue generation.

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6. Geographic Revenue Analysis

Revenue analysis by state identified significant regional differences.

Top Revenue State

São Paulo (SP) generated the highest revenue by a substantial margin.

Other strong-performing states include:

- Rio de Janeiro (RJ)
- Minas Gerais (MG)

Business Interpretation

These regions represent the most valuable customer markets and may warrant additional marketing investment and customer retention efforts.

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7. Customer Segmentation Potential

The engineered feature set contains strong signals for future customer segmentation.

Relevant segmentation features include:

- Recency
- Frequency
- Monetary Value
- Review Behavior
- Product Diversity
- Seller Diversity
- Delivery Performance

Potential customer groups may include:

- Champions
- Loyal Customers
- Premium Customers
- At-Risk Customers
- Dormant Customers
- One-Time Buyers

8. Recommendations for Machine Learning

Before clustering and predictive modeling, the following preprocessing steps are recommended:

1. Impute missing delivery features.
2. Apply log transformation to monetary value.
3. Standardize numerical variables.
4. Remove highly correlated features where necessary.
5. Perform PCA for dimensionality reduction.
6. Conduct customer segmentation using K-Means, DBSCAN, and Gaussian Mixture Models.

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9. Conclusion

The customer feature warehouse successfully transformed raw transactional data into meaningful business-oriented customer features.

The analysis revealed:

- A highly skewed spending distribution.
- Significant customer value variation.
- Strong regional revenue concentration.
- Limited missing data.
- Clear opportunities for customer segmentation.

The engineered dataset is now suitable for advanced machine learning applications including clustering, churn prediction, customer lifetime value estimation, anomaly detection, and explainable AI.